Q1.

A dairy industry researcher, Robyn, decided to investigate the milk yield, classified as low, medium or high, obtained from four different breeds of cow, A, B, C and D.

The milk yield of a sample of 105 cows was monitored and the results are summarised in contingency **Table 1**.

Table 1

		Yield			
		Low	**Medium**	**High**	**Total**
Breed	**A**	4	5	12	**21**
	B	10	6	4	**20**
	C	8	17	7	**32**
	D	5	20	7	**32**
	Total	**27**	**48**	**30**	**105**

The sample of cows may be regarded as random.

Robyn decides to carry out a χ^2-test for association between milk yield and breed using the information given in **Table 1**.

(a) Contingency **Table 2** gives some of the expected frequencies for this test.

Complete **Table 2** with the missing expected values.

Table 2

		Yield		
		Low	**Medium**	**High**
Breed	**A**			6
	B	5.14	9.14	5.71
	C			
	D	8.23	14.63	9.14

(2)

(b) (i) For Robyn's test, the test statistic $\sum \frac{(O-E)^2}{E} = 19.4$ correct to three significant figures.

Use this information to carry out Robyn's test, using the 1% level of significance.

(5)

(ii) By considering the observed frequencies given in **Table 1** with the expected frequencies in **Table 2**, interpret, in context, the association, if any, between milk yield and breed.

(2)

(Total 9 marks)

Q2.

A large estate agency would like all the properties that it handles to be sold within three months. A manager wants to know whether the type of property affects the time taken to sell it. The data for a random sample of properties sold are tabulated below.

	Type of property				
	Flat	**Terraced**	**Semi-detached**	**Detached**	**Total**
Sold within three months	4	34	28	18	84
Sold in more than three months	9	18	8	6	41
Total	13	52	36	24	125

(a) Conduct a χ^2-test, at the 10% level of significance, to determine whether there is an association between the type of property and the time taken to sell it. Explain why it is necessary to combine two columns before carrying out this test.

(10)

(b) The manager plans to spend extra money on advertising for one type of property in an attempt to increase the number sold within three months. Explain why the manager might choose:

(i) terraced properties;

(ii) flats.

(2)
(Total 12 marks)

Q3.

Fiona, a lecturer in a school of engineering, believes that there is an association between the class of degree obtained by her students and the grades that they had achieved in A-level Mathematics.

In order to investigate her belief, she collected the relevant data on the performances of a random sample of 200 recent graduates who had achieved grades A or B in A-level Mathematics. These data are tabulated below.

		Class of degree				
		1	**2(i)**	**2(ii)**	**3**	**Total**
A-level grade	**A**	20	36	22	2	80
	B	9	55	48	8	120

Total	29	91	70	10	200

(a) Conduct a χ^2 test, at the 1% level of significance, to determine whether Fiona's belief is justified.

(9)

(b) Make **two** comments on the degree performance of those students in this sample who achieved a grade B in A-level Mathematics.

(2)
(Total 11 marks)

Q4.

Emily believed that the performances of 16-year-old students in their GCSEs are associated with the schools that they attend. To investigate her belief, Emily collected data on the GCSE results for 2010 from four schools in her area.

The table shows Emily's collected data, denoted by O_i, together with the corresponding expected frequencies, E_i, necessary for a χ^2 test.

	≥ 5 GCSEs		1 ≤ GCSEs < 5		No GCSEs	
	O_i	E_i	O_i	E_i	O_i	E_i
Jolliffe College for the Arts	187	193.15	93	90.62	30	26.23
Volpe Science Academy	175	184.43	97	86.52	24	25.05
Radok Music School	183	183.81	78	86.23	34	24.96
Bailey Language School	265	248.61	112	116.63	22	33.76

Emily used these values to correctly conduct a χ^2 test at the 1% level of significance.

(a) State the null hypothesis that Emily used.

(1)

(b) Find the value of the test statistic, χ^2, giving your answer to one decimal place.

(3)

(c) State, in context, the conclusion that Emily should reach based on the results of her χ^2 test.

(3)

(d) Make one comment on the GCSE performances of 16-year-old students attending Bailey Language School.

(1)

(e) Emily's friend, Joanna, used the same data to correctly conduct a χ^2 test using the 10% level of significance.

State, with justification, the conclusion that Joanna should reach.

(2)
(Total 10 marks)

Q5.

Julie, a driving instructor, believes that the first-time performances of her students in their driving tests are associated with their ages.

Julie's records of her students' first-time performances in their driving tests are shown in the table.

Age	Pass	Fail
17 – 18	28	20
19 – 30	2	14
31 – 39	12	33
40 – 60	6	5

(a) Use a χ^2 test at the 1% level of significance to investigate Julie's belief.

(9)

(b) Interpret your result in part (a) as it relates to the 17 – 18 age group.

(1)
(Total 10 marks)

Q6.

It is thought that the incidence of asthma in children is associated with the volume of traffic in the area where they live.

Two surveys of children were conducted: one in an area where the volume of traffic was heavy and the other in an area where the volume of traffic was light.

For each area, the table shows the number of children in the survey who had asthma and the number who did not have asthma.

	Asthma	No asthma	Total
Heavy traffic	52	58	110
Light traffic	28	62	90
Total	80	120	200

(a) Use a χ^2 test, at the 5% level of significance, to determine whether the incidence of asthma in children is associated with the volume of traffic in the area where they live.

(8)

(b) Comment on the number of children in the survey who had asthma, given that they lived in an area where the volume of traffic was heavy.

(1)
(Total 9 marks)

Q7.

A statistics unit is required to determine whether or not there is an association between students' performances in mathematics at Key Stage 3 and at GCE.

A survey of the results of 500 students showed the following information:

		GCE Grade				
		A	**B**	**C**	**Below C**	**Total**
Key Stage 3 Level	**8**	60	55	47	43	205
	7	55	32	31	26	144
	6	40	38	35	38	151
	Total	155	125	113	107	500

(a) Use a χ^2 test at the 10% level of significance to determine whether there is an association between students' performances in mathematics at Key Stage 3 and at GCE.

(9)

(b) Comment on the number of students who gained a grade A at GCE having gained a level 7 at Key Stage 3.

(1)
(Total 10 marks)

Q8.

Year 12 students at Newstatus School choose to participate in one of four sports during the Spring term.

The students' choices are summarised in the table.

	Squash	**Badminton**	**Archery**	**Hockey**	**Total**
Male	5	16	30	19	70
Female	4	20	33	53	110
Total	9	36	63	72	180

(a) Use a χ^2 test, at the 5% level of significance, to determine whether the choice of sport is independent of gender.

(10)

(b) Interpret your result in part (a) as it relates to students choosing hockey.

(2)

(Total 12 marks)

Q9.

It is claimed that the area within which a school is situated affects the age profile of the staff employed at that school. In order to investigate this claim, the age profiles of staff employed at two schools with similar academic achievements are compared.

Academia High School, situated in a rural community, employs 120 staff whilst Best Manor Grammar School, situated in an inner-city community, employs 80 staff.

The **percentage** of staff within each age group, for each school, is given in the table.

Age	Academia High School	Best Manor Grammar School
22-34	17.5	40.0
35-39	60.0	45.0
40-59	22.5	15.0

(a) (i) Form the data into a contingency table suitable for analysis using a χ^2 distribution.

(2)

(ii) Use a χ^2 test, at the 1% level of significance, to determine whether there is an association between the age profile of the staff employed and the area within which the school is situated.

(9)

(b) Interpret your result in part (a)(ii) as it relates to the 22-34 age group.

(2)

(Total 13 marks)

Q10.

A political party held a conference, attended by delegates from local branches, to choose a new leader. The voting at the conference was carried out in public. In a confidential survey, randomly selected delegates were asked whether they would have voted differently if voting had been by secret ballot. The answers, classified by the ages of the delegates, are summarised in the following table.

	Definitely	Possibly	Definitely not
Under 65 years	**4**	**12**	6
65 years and over	**10**	**28**	54

(a) Use a χ^2 distribution and the 5% significance level to examine whether the answer to the question is associated with the age of the delegate.

(10)

(b) Interpret your result in part (a) in the context of the question.

(1)

(c) Make a further comment about the ages of the delegates.

(1)
(Total 12 marks)

Q11.

Chandra, an estate agent, sells properties in a large conurbation. She classifies the areas of the conurbation as:

City Centre;
Inner City – close to but outside the City Centre;
Suburban – outside the Inner City.

The following table shows the number of properties she has for sale in each of these areas together with information on the Advertised Price.

		Area		
		City Centre	**Inner City**	**Suburban**
Advertised Price	**< £120 000**	2	58	20
	≤ £120 000	34	12	44

(a) Use a χ^2 distribution and the 1% significance level to analyse this contingency table.

(9)

(b) Interpret your result in the context of this question.

(2)
(Total 11 marks)

Q12.

Rhamesh runs a café which opens at noon and admits no further customers after 9 pm. He does not sell alcoholic drinks, but customers can bring their own beer or wine with them, to drink with their meal, if they wish.

To help him decide whether it would be worthwhile selling his own alcoholic drinks, Rhamesh asks Anna, a student, to record how many customers bring their own alcoholic drinks. Anna suspects this may be related to the time of day, and following her observations she produces the table below.

The table shows the number of customers, with and without alcoholic drinks, who entered the café at different times on a particular day.

	Time of Day		
	Noon to 3 pm	**3 pm to 6 pm**	**6 pm to 9 pm**
Alcoholic drink	12	6	54
No alcoholic drink	62	35	67

(a) Use the 1% significance level to investigate whether taking alcohol with a meal is independent of time of day.

(9)

(b) (i) Interpret your result in part (a) as it relates to customers entering the cafe between 6 pm and 9 pm.

(ii) Make a further statement about customers entering the caf6 between 6 pm and 9 pm.

(3)

(c) How might your conclusions be affected, if at all, by the fact that the table shows the customers who entered the cafe on only one particular day?

(2)

(Total 14 marks)

Q13.

A well known picture of the Beatles shows them on a pedestrian crossing outside the Abbey Road recording studios. Substantial numbers of Beatles fans visit Abbey Road and use the pedestrian crossing.

It was claimed that these fans were causing delays to rush hour traffic. As part of an investigation, people using the crossing were asked whether they were using it in the course of their normal daily lives or because of the Beatles photograph. The answers and the times of day are summarised in the contingency table below.

	Time of day	
	Rush hour	**Out of rush hour**
Normal daily lives	34	45
Because of Beatles	4	28

(a) Using the 5% significance level, examine whether the reason for using the crossing is associated with the time of day.

(10)

(b) Interpret your result in the context of this question.

(1)

(Total 11 marks)

Q14.

Students following a media studies course at a particular university must study a foreign language. They are given a choice of Japanese, Russian or Swedish. The following table summarises the choices by gender of a random sample of students entering the course.

	Japanese	Russian	Swedish
Female	16	12	22
Male	12	14	54

(a) Use a χ^2 distribution and the 5% significance level to investigate whether choice of language is associated with gender.

(10)

(b) (i) Interpret your result in part (a) as it relates to the choice of Swedish.

(ii) Make a further statement about the choice of Swedish.

(3)
(Total 13 marks)

Q15.

A railway company wishes to monitor passenger opinion. During a pilot study a randomly selected sample of rail passengers was surveyed. As part of the survey they were asked whether they carried mobile phones for use on their journey and whether they were irritated by other passengers using mobile phones on the journey.

The results are summarised in the following table.

	Irritated	Not irritated
Mobile phone carried	12	14
Mobile phone not carried	26	6

(a) (i) Use a χ^2 test, at the 5% significance level, to analyse the contingency table above.

(10)

(ii) Interpret your conclusion in the context of the question.

(1)

(b) As a result of the pilot survey the questionnaire was modified and a further random sample of rail passengers was asked to complete it. The answers to the questions relating to mobile phones are summarised in the following table.

	Not irritated	Slightly irritated	Irritated	Extremely irritated
Mobile phone carried	6	46	18	3
Mobile phone not carried	3	7	14	17

The expected values for the analysis of this contingency table are given below.

	Not irritated	Slightly irritated	Irritated	Extremely irritated
Mobile phone carried	5.76	33.94	20.49	12.81
Mobile phone not carried	3.24	19.06	11.51	7.19

Jason is asked to analyse the data and suggests the “Not irritated” and “Extremely irritated” columns be combined together before carrying out the analysis.

Eric suggests that it would be better to combine the “Not irritated” column with the “Slightly irritated” column and the “Extremely irritated” column with the “Irritated” column.

Anthony suggests that the “Not irritated” column should be combined with the “Slightly irritated” column and the resulting 2×3 table analysed.

(i) Give one criticism of Jason’s suggestion.

(1)

(ii) Explain why Anthony’s suggestion is preferable to Eric’s.

(2)

(iii) Carry out Anthony’s suggestion, using the 1% significance level, and interpret your conclusion in context.

(8)

(Total 22 marks)